Gaurav Ghosh

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Professional Summary

Computer Science undergraduate with experience building backend, Generative AI, machine-learning, and GPU-accelerated systems using Python, FastAPI, Spring Boot, Docker, RAG, and vector databases. Strong problem-solving foundation demonstrated through 1000+ algorithmic problems, LeetCode Guardian, and Codeforces Expert ratings.

Education

National Institute of Technology, Silchar — **Assam, India**
Bachelor of Technology in Computer Science and Engineering; CGPA: 9.15 — *Aug 2023 – Expected 2027*
Class XII, CBSE: 93.2% | **Class X, ICSE:** 96.8% — *2021 – 2023*

Experience & Research

S.N. Bose National Centre for Basic Sciences — **Silchar, India**
Research Intern – GPU-Accelerated Voronoi Diagram Generation — *May 2026 – Present*

- Implemented a CPU/GPU-based digital circle-growing Voronoi construction pipeline using **Python**, **NumPy**, and **Taichi** for large two-dimensional grids.
- Added **Cutoff Radius pruning** and **Parallel Reduction** techniques to reduce redundant computations in geometry-processing workloads.
- Developing benchmarks against brute-force and Jump Flooding Algorithm baselines to compare runtime and output correctness across different grid and site configurations.

Technical Skills

Languages: Python, Java, C++, SQL, JavaScript, C

Backend Development: FastAPI, Spring Boot, REST APIs, WebSockets, Streamlit

Generative AI & Machine Learning: RAG, LangChain, Hugging Face, Qdrant, Semantic Embeddings, Vector Search, LLMs, PyTorch, scikit-learn, NLP

Databases & Developer Tools: MySQL, MongoDB, Docker, Git, GitHub, Linux, Postman

Computer Science Fundamentals: Data Structures and Algorithms, OOP, Operating Systems, DBMS, Computer Networks

Projects

Sandboxed Multilanguage Online Compiler [Live Demo] — *Backend System — 2026*

- Built a code-execution platform with a **Spring Boot** backend supporting real-time compilation and execution of user-submitted **C++, Java, and Python** programs.
- Isolated code submissions inside dedicated **Docker containers** with CPU, memory, and network restrictions to reduce risks associated with executing untrusted programs.
- Returned compilation output, execution results, and runtime failures from the backend to provide structured feedback for submitted programs.

Enterprise Policy Analyzer [GitHub] — *Generative AI Application — 2026*

- Built a multi-document **Retrieval-Augmented Generation** application for answering questions across unstructured corporate policies and contracts.
- Processed PDFs using **pdfplumber** and LangChain text splitters, generated semantic embeddings using **Hugging Face**, and stored vectors in **Qdrant** for similarity retrieval.
- Integrated the **Groq LLM API** to generate grounded answers containing source filenames and page-number references, along with workflows for summarization, clause extraction, risk identification, and metadata tagging.

TrafficOps AI: Command Center [GitHub] | [Live Demo] — *AI/ML Application — 2026*

- Developed a **FastAPI** prediction service using trained scikit-learn models to estimate road-closure risk, impact severity, and field-resource requirements from spatial, temporal, and historical features.
- Implemented an adaptive feedback-learning mechanism that updates deployment recommendations using post-event outcomes and validated its behaviour over **10 simulation cycles**.
- Integrated MapmyIndia and Leaflet to generate and visualize graph-based diversion routes around blocked junctions.

Achievements

- **Competitive Programming:** LeetCode Guardian with peak rating **2235**, Codeforces Expert with peak rating **1663**, AtCoder Green, and 1000+ algorithmic problems solved.
- Secured **Rank 453** in Codeforces Round 1095 and ranked in the top **4.6%** among 34,700+ teams in Flipkart GridLock Hackathon 2.0.
- Achieved a perfect **10.00 SGPA** during the third semester at NIT Silchar.

Leadership

Marketing Head, Esports Club – NIT Silchar — *2025 – Present*

- Lead a 15-member team managing event operations, promotional campaigns, sponsor outreach, and corporate partnerships for campus-wide competitive events.